STASIS-SCAN

A decision explorer for surgical planning in early lung cancer

SARA HABIBI
SARAA.HABIBI@MAIL.UTORONTO.CA

Lung Cancer: A Major Global Health Challenge

Lung cancer remains the leading cause of cancer related mortality worldwide

Global burden

- ~1.8 million deaths annually worldwide
- One of the most frequently diagnosed malignancies globally

Canadian context

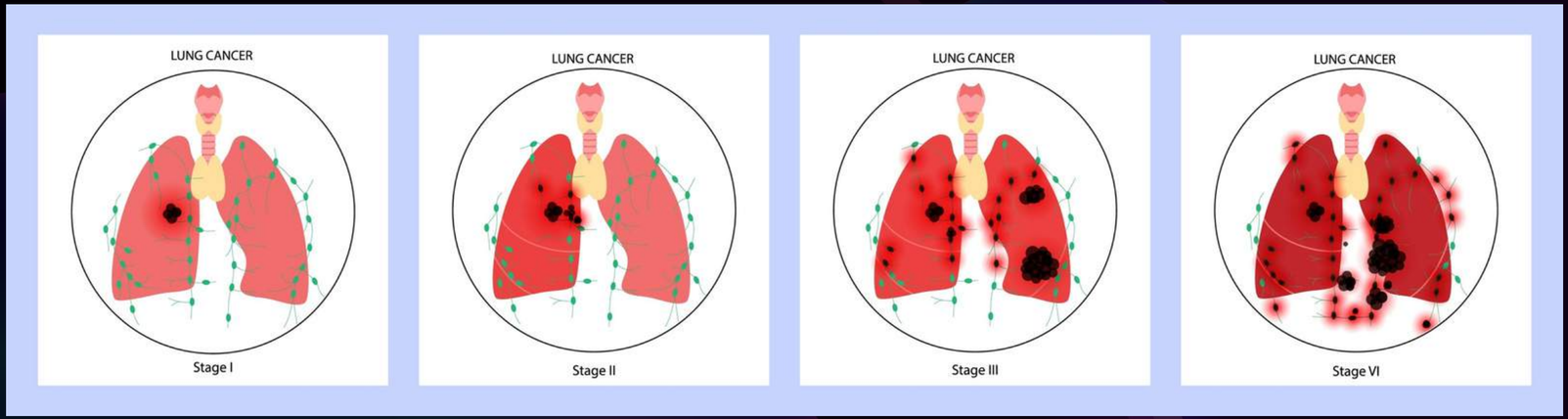
- Over 30,000 new diagnoses each year
- Roughly 60 percent present with early stage disease

Clinical implication

Early stage lung cancer is commonly treated with surgical resection

STAS: SPREAD THROUGH AIR SPACES

Spread Through Air Spaces (STAS) is a distinct pattern of tumor invasion in lung cancer that was formally recognized by the World Health Organization in 2015. It refers to the presence of tumor cells within the alveolar air spaces beyond the main tumor margin, where clusters of cancer cells detach from the primary tumor and disperse through the surrounding lung tissue. Under microscopic examination, these cells can appear as micropapillary clusters, solid nests, or individual tumor cells within alveolar spaces adjacent to the tumor. Unlike traditional invasion pathways such as vascular or lymphatic spread, STAS represents a unique mechanism of local tumor dissemination through the lung's air space architecture.



WHY STAS MATTERS CLINICALLY?

STAS is relatively common in early stage lung cancer

Prevalence

- Present in approximately 16 to 27 percent of early stage lung tumors
- Observed across multiple histologic subtypes of lung adenocarcinoma

Clinical impact

- Associated with 2 to 4 times higher risk of local recurrence
- Linked to more aggressive tumor behavior

Interaction with surgical strategy

- Limited resections may fail to remove dispersed tumor clusters
- STAS positive tumors show worse outcomes after sublobar resection

Clinical observation

STAS positive patients undergoing segmentectomy show higher recurrence and mortality compared with lobectomy

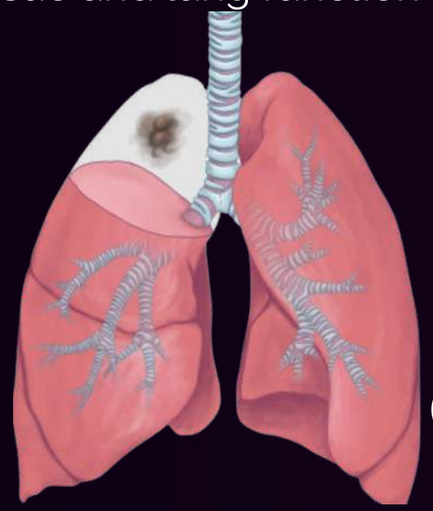
SURGICAL TREATMENT OPTIONS

Lobectomy

Removal of an entire lung lobe
Historically considered the oncologic standard

Segmentectomy

Removal of the tumor containing lung segment
Preserves more lung tissue and lung function



Selecting the optimal procedure for each patient

MOLECULAR BIOLOGY OF STAS

STAS is not a random phenomenon it reflects underlying tumor biology and cellular behavior

Many STAS positive tumors show activation of epithelial mesenchymal transition (EMT)

EMT is a biological program in which epithelial cells lose adhesion and gain migratory capacity.

Key molecular biomarkers associated with STAS

- Loss of E cadherin

  Reduced cell adhesion allowing tumor cells to detach from the primary tumor

- Gain of N cadherin

  Promotes cellular mobility and invasive behavior

- TP53 mutation enrichment

  Frequently observed in STAS positive tumors and linked to aggressive tumor biology

- ALK rearrangements and PTEN alterations

  Associated with enhanced tumor progression and invasive potential

These molecular changes promote tumor cell detachment, migration, and dispersion through alveolar air spaces, enabling the STAS invasion pattern.

STASIS-SCAN

STASis-Scan is an interactive decision exploration platform designed to evaluate recurrence risk in early stage lung cancer.

The system integrates clinical features, tumor characteristics, and molecular biomarkers associated with STAS.

Interactive Decision making tool

Model inputs

- Clinical variables such as age
- Tumor characteristics including tumor size
- Molecular mutations such as KRAS, TP53, and EGFR
- EMT related biomarkers including E cadherin and N cadherin
- Surgical procedure type

Model outputs

- Predicted five year recurrence risk
- Comparison of lobectomy vs segmentectomy outcomes
- Visualization of how tumor biology influences surgical risk

Goal

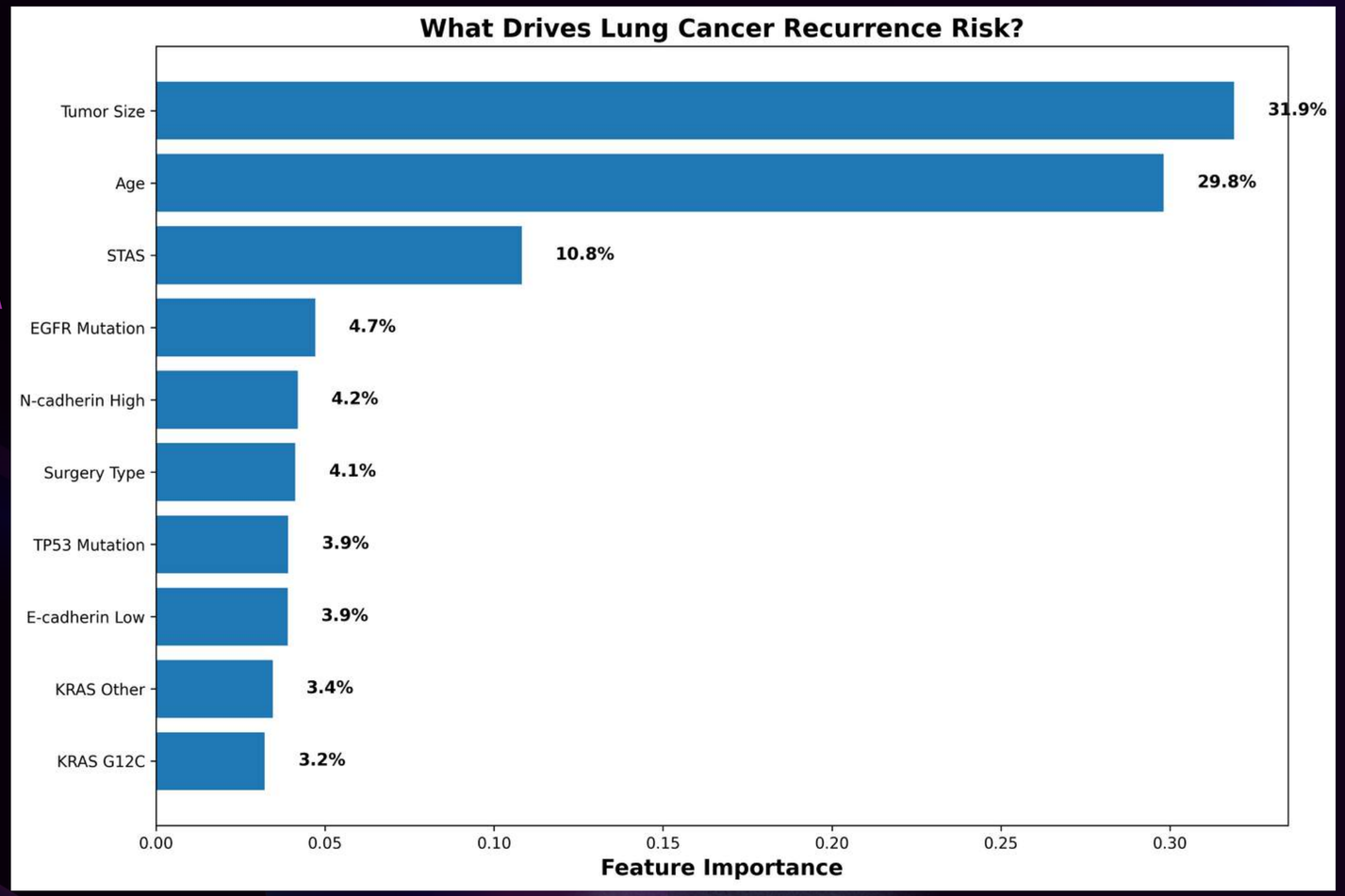
Translate published research on STAS and tumor biology into a simple and interpretable decision exploration tool that supports surgical planning.

DEMONSTRATION OF THE MODEL

HTTPS://STASIS-SCAN-AI-SH.STREAMLIT.APP/

MODEL FRAMEWORK

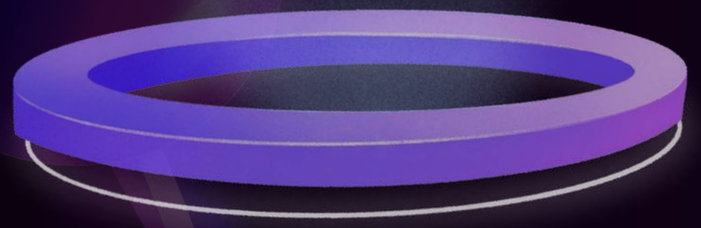
The STASis Scan model integrates clinical variables, tumor characteristics, and molecular biomarkers to estimate lung cancer recurrence risk. The feature importance analysis shown here highlights tumor size, patient age, and STAS status as major contributors to the prediction, while molecular markers and surgical variables provide additional refinement. These factors are combined using a probabilistic model to generate an interpretable estimate of five year recurrence risk.



MODEL FRAMEWORK EQUATION APPROXIMATION

Approximate Risk Equation (Linear Representation)

Risk≈0.040+0.198(STAS)+0.019(Tumor Size)+0.046(TP53)+0.042(Surgery)Risk \approx 0.040 + 0.198(STAS) + 0.019(Tumor\ Size) + 0.046(TP53) + 0.042(Surgery)Risk≈0.040+0.198(STAS)+0.019(Tumor Size)+0.046(TP53)+0.042(Surgery)Note: This equation is a simplified linear approximation for interpretability. The primary predictive model is a Random Forest classifier with 83% accuracy.



MODEL PERFORMANCE

Model Type	Random Forest classifier
Performance Metric	Accuracy: 83% AUC: 0.80
Literature Design	Synthetic dataset calibrated to published literature Relationships based on reported effect sizes from STAS research studies.
Purpose	Research prototype for exploring how tumor biology may influence surgical risk.

KEY TAKEAWAYS

STAS is an important predictor of lung cancer recurrence.
Current surgical decisions often occur without knowledge of STAS status.
Integrating tumor biology with clinical data can improve risk awareness during surgical planning.
STASis Scan translates research insights into an interpretable decision exploration tool.
Mission
Make STAS part of the surgical conversation before the incision.

THANK YOU!

Much thanks to the developer → Sara Habibi